$$Z_Q(s) = \sum_{a \in \mathcal{A}} \exp Q(s, a)$$

$$\mathcal{C}_H(Q) = \mathbb{E}_{s \sim \mathcal{D}}\Big[\log Z_Q(s) - \mathbb{E}_{a \sim \widehat{\pi}_\beta(\cdot|s)} Q(s, a)\Big]$$

$$\widehat{Q}^{k+1} \leftarrow \arg\min_Q \left[\mathcal{L}_B(Q) + \alpha \mathcal{C}_H(Q)\right]$$